

Topology-Locked Flow Organization in a Rotationally-Stacked Packed Bed: Pore-Scale Lattice-Boltzmann Simulation Validated by PIV

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1 Introduction

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2 Geometry and Benchmark Configuration

The experimental reference configuration is described in detail by velten2026. The packing consists of polyhedral bar modules, each containing five parallel bars of height $l_{\text{ch}} = 10$ mm separated by slits of width $l_P = 5$ mm, housed within a circular cross-section of diameter $d_{\text{inlet}} = 62.12$ mm (Fig. ??, reproduced from velten2026). The void-volume fraction (porosity) is $\varphi = 0.332$. Modules are 3D-printed from PA 12 via laser sintering and can be rotated against each other in increments of 5° via 37 indexing slits on the rim velten2026.

2.1 Twist-and-Repeat Topology

Write §2.1 – twist-and-repeat, N=6 for 30 deg, N=9 for 40 deg

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2.2 Three Topologies Studied

The particle Reynolds number is defined following velten2026 as

$$Re_P = \frac{v_s \rho l_{ch}}{\varphi \mu}, \quad (1)$$

where v_s is the superficial velocity, $\rho = 1.204 \text{ kg/m}^3$ and $\mu = 1.822 \times 10^{-5} \text{ kg/(m s)}$ are the density and dynamic viscosity of air at $T = 20^\circ \text{C}$ and $p = 1 \text{ atm}$, and $l_{ch} = 10 \text{ mm}$ is the bar width velten2026.

2.3 Measurement Positions and Coordinate System

Write §2.3 – coordinate system and measurement positions

3 Numerical Methodology, Verification and Validation

3.1 Lattice Boltzmann Formulation (ALBORZ)

To model the flow field we make use in this study of the lattice Boltzmann method. The lattice Boltzmann method is a solver for the Boltzmann equation in the limit of the hydrodynamic regime [?]. It can be derived by first discretizing the Boltzmann equation in phase-space, the space of particle degrees of freedom such as translational, via a finite-order spectral approach. The Hermite polynomials are typically the global shape functions used for that purpose. The resulting system of coupled hyperbolic equations is then integrated along characteristics [?]. This in turn yields the now-famous “stream-collide” equation:

$$f_i(\mathbf{r} + \mathbf{c}_i \delta t, t + \delta t) - f_i(\mathbf{r}, t) = \Omega_i, \quad (2)$$

for discrete populations f_i where $\mathbf{c}_i$ are discrete particle velocities corresponding to the populations, Ω_i is the discrete collision operator modeling equilibration of the distribution function due to molecular collisions and δt is the time-step size. Close to local thermodynamic equilibrium (LTE) conditions (which is valid for small Knudsen numbers, as found in the flows of interest) a linear relaxation operator also known as Bhatnagar-Gross-Krook (BGK) is usually employed, which in its discrete form is:

$$\Omega_i = \omega (f_i^{\text{eq}} - f_i) + \Xi_i, \quad (3)$$

where ω is the relaxation frequency coefficient and f_i^{eq} is the discrete form of the continuous attractor obtained via minimization of entropy, i.e. the Maxwell-Boltzmann distribution [?,?]. It must be noted that here the collision operator is supplemented with a term Ξ_i accounting for deviations in the diagonal components of the third-order moments of the equilibrium term (for models relying on third-order quadratures). In the context of the lattice

Boltzmann method the discrete equilibrium distribution function is expressed as a finite-order expansion of the Maxwell-Boltzmann distribution in terms of Hermite polynomials by means of the Gauss-Hermite quadrature:

$$f_i^{\text{eq}} = w_i \sum_{n=0}^N \frac{1}{n! c_s^{2n}} \mathcal{H}_n(\mathbf{c}_i) : a_n^{\text{eq}}(\rho, \mathbf{u}, p/\rho), \quad (4)$$

where $\mathcal{H}_n$ is the Hermite polynomial tensor of order/rank n , a_n^{eq} the corresponding equilibrium coefficient, w_i are weights associated with the quadrature and c_s the lattice speed of sound also tied to the quadrature. ρ is the fluid density, $\mathbf{u}$ the velocity, p the pressure and N the order of expansion. For the remainder of the manuscript we will restrict ourselves to the D2Q9 and D3Q27 stencils, i.e. third-order quadrature, where D is the physical dimension and Q the number of discrete velocities. In 2-D N is set to four while in 3-D it is set to six. Both are dictated by the maximum number of terms supported by the stencils. Contrary to classical second-order polynomial expansions, the present model allows the discrete form to preserve Galilean-invariance of the dissipation rate of shear modes at the Navier-Stokes level. The term Ξ_i restores Galilean-invariance to the dissipation rate of normal modes and is defined as [?,?]:

$$\Xi_i = \left(1 - \frac{\omega}{2}\right) \frac{w_i}{2c_s^4} (\nabla \mathcal{H}_2) : \Delta_3^{\text{eq}}, \quad (5)$$

where Δ_3^{eq} is a diagonal tensor of rank three defined as:

$$\Delta_{\alpha\alpha\alpha}^{\text{eq}} = \rho u_\alpha (u_\alpha^2 + 3p/\rho) - \sum_i c_{i,\alpha}^3 f_i^{\text{eq}}. \quad (6)$$

More advanced versions of the BGK operator are usually employed on the basis of physical (variable Prandtl number) or numerical arguments (stability). Among those the multiple relaxation time formulation allows to relax the distribution function in a space of linearly independent base functions, typically moments of the distribution function. Owing to its improved stability/accuracy it has become quite popular in the lattice Boltzmann community. Here, we make use of a multiple relaxation time operator in a space of modified Hermite central moments allowing for independent control over the bulk viscosity [?]. At the difference of the classical Hermite polynomial space [?,?], the trace and trace-free contributions to the second-order moments relax independently in this formulation:

$$\Omega_i = \mathcal{T}^{-1} W \mathcal{T} (f_i^{\text{eq}} - f_i) + \Xi_i \quad (7)$$

where $\mathcal{T}$ and $\mathcal{T}^{-1}$ are the moments transform tensor and its inverse and W is the diagonal tensor of relaxation rates. The moments transform tensor is defined via a set of modified central Hermite polynomials $\tilde{\mathcal{H}}_n(\mathbf{c}_i)$, which for

the D2Q9 stencil are:

$$\tilde{\mathcal{H}}_0 = 1, \quad (8a)$$

$$\tilde{\mathcal{H}}_x = c_{i,x} - u_x, \quad (8b)$$

$$\tilde{\mathcal{H}}_y = c_{i,y} - u_y, \quad (8c)$$

$$\tilde{\mathcal{H}}_{xy} = (c_{i,x} - u_x)(c_{i,y} - u_y), \quad (8d)$$

$$\tilde{\mathcal{H}}_{xx-yy} = (c_{i,x} - u_x)^2 - (c_{i,y} - u_y)^2, \quad (8e)$$

$$\tilde{\mathcal{H}}_{xx+yy} = (c_{i,x} - u_x)^2 + (c_{i,y} - u_y)^2 - 2c_s^2, \quad (8f)$$

$$\tilde{\mathcal{H}}_{xxy} = (c_{i,x} - u_x)^2(c_{i,y} - u_y) - c_s^2(c_{i,y} - u_y), \quad (8g)$$

$$\tilde{\mathcal{H}}_{xyy} = (c_{i,y} - u_y)^2(c_{i,x} - u_x) - c_s^2(c_{i,x} - u_x), \quad (8h)$$

$$\tilde{\mathcal{H}}_{xxyy} = \left[(c_{i,x} - u_x)^2 - c_s^2\right] \left[(c_{i,y} - u_y)^2 - c_s^2\right]. \quad (8i)$$

Equilibrium moments based on these polynomials are the central Hermite coefficients, $\tilde{a}_n^{\text{eq}}$ which for the equilibrium distribution function (for the D2Q9) lead to:

$$\tilde{a}_0^{\text{eq}} = \rho, \quad (9a)$$

$$\tilde{a}_x^{\text{eq}} = 0, \quad (9b)$$

$$\tilde{a}_y^{\text{eq}} = 0, \quad (9c)$$

$$\tilde{a}_{xy}^{\text{eq}} = 0, \quad (9d)$$

$$\tilde{a}_{xx-yy}^{\text{eq}} = 0, \quad (9e)$$

$$\tilde{a}_{xx+yy}^{\text{eq}} = 2(p - \rho c_s^2), \quad (9f)$$

$$\tilde{a}_{xxy}^{\text{eq}} = 0, \quad (9g)$$

$$\tilde{a}_{xyy}^{\text{eq}} = 0, \quad (9h)$$

$$\tilde{a}_{xxyy}^{\text{eq}} = \rho(p/\rho - c_s^2)^2. \quad (9i)$$

Setting $p = \rho c_s^2$ all equilibrium coefficients reduce to zero except at order zero. The relaxation frequencies tensor W for the D2Q9 stencil is defined as:

$$W = \text{diag}(1, 1, 1, \omega_s, \omega_s, \omega_b, \omega_g, \omega_g, \omega_g), \quad (10)$$

where the operator diag is defined as:

$$\text{diag}(\mathbf{A}) = (\mathbf{A} \otimes \mathbf{1}) \circ \mathbf{I}, \quad (11)$$

with $\mathbf{A}$ a given vector, $\mathbf{1}$ a vector with elements 1, $\mathbf{I}$ the unitary tensor and $\circ$ the Hadamard product. ω_s , ω_b and ω_g are the shear, bulk and ghost modes relaxation frequencies respectively. ω_b is related to the bulk viscosity, η , as:

$$\omega_b = \frac{\delta t}{\left(\frac{2+D}{D} - \frac{\partial \ln p}{\partial \ln \rho}\right) \frac{\eta}{p} + \delta t/2}. \quad (12)$$

The correction term for the diagonal components of the third-order moments appearing in Eq. ?? changes in the context of this new solver into:

$$\Xi_i = \left(1 - \frac{\omega_b \omega_s}{\omega_b + \omega_s}\right) \frac{w_i}{2c_s^4} (\nabla \mathcal{H}_2) : \Delta_3^{\text{eq}}. \quad (13)$$

The lattice Boltzmann model as introduced here allows for an independent bulk viscosity and Galilean-invariant dynamic and bulk viscosities.

3.2 Computational Domain and Boundary Conditions

While a variety of different boundary conditions have been developed for the lattice Boltzmann method, in the context of the present study all solid boundaries are modeled using the half-way bounce-back scheme where missing populations after the collision-streaming steps are computed as [?]:

$$f_\alpha(\mathbf{x}, t + \delta_t) = f_{\bar{\alpha}}^*(\mathbf{x}, t), \quad (14)$$

where f^* is the post-collision population (prior to streaming) and $\bar{\alpha}$ is the index of the particle velocity opposite that of α . To get rid of the stair-case approximation the half-way bounce-back is supplemented with the treatment for curved boundaries as proposed in [?].

At a given boundary node $\mathbf{x}_f$, the missing incoming populations are computed as:

$$f_\alpha(\mathbf{x}_f, t + \delta_t) = 2q f_{\bar{\alpha}}(\mathbf{x}_f + \mathbf{c}_{\bar{\alpha}}, t + \delta_t) + (1 - 2q) f_{\bar{\alpha}}(\mathbf{x}_f, t + \delta_t), \quad \forall q < \frac{1}{2}, \quad (15a)$$

$$f_\alpha(\mathbf{x}_f, t + \delta_t) = \frac{1}{2q} f_{\bar{\alpha}}(\mathbf{x}_f + \mathbf{c}_{\bar{\alpha}}, t + \delta_t) + \frac{2q - 1}{2q} f_{\bar{\alpha}}(\mathbf{x}_f, t + \delta_t), \quad \forall q \geq \frac{1}{2}, \quad (15b)$$

where $\bar{\alpha}$ designates the direction opposite α and q :

$$q = \frac{\|\mathbf{x}_f - \mathbf{x}_s\|}{\|\mathbf{c}_\alpha\|}, \quad (16)$$

with $\mathbf{x}_s$ denoting the wall position in direction α .

Other boundary conditions applied at the inlets and outlets, i.e. constant velocity and constant pressure, are realized using the non-equilibrium extrapolation approach proposed in [?]. In this approach the missing populations at the boundary nodes are reconstructed as:

$$f_\alpha(\mathbf{x}_w, t + \delta_t) = f_\alpha^{(eq)}(\mathbf{x}_w, t + \delta_t) + f_\alpha^{(neq)}(\mathbf{x}_w, t + \delta_t), \quad (17)$$

where $\mathbf{x}_w$ is the position of the boundary node and $f^{(neq)}$ the non-equilibrium part of the distribution function. One of the two variables (density or velocity) is known via the boundary condition and the other is computed by interpolation from neighboring nodes. The non-equilibrium part of the distribution function is also computed via an interpolation in space.

3.3 Simulation Parameters and Case Matrix

Write §3.3 – simulation parameters; fill collision model/stencil, cores, convergence

3.4 Grid Independence Study

Write §3.4 – grid independence; fill table numbers

3.5 Symmetry and Consistency Verification

The antisymmetry of the locked mean flow about the column axis (Pos 3 vs Pos 5-flipped) provides both a numerical consistency check and a physical verification of the 180° -symmetry of the slit pattern. Fig. ?? shows the normalised velocity-magnitude residual $|\Delta V|/U_{\text{ref}}$ across the measurement plane; the mean residual is 0.15% and the maximum is 0.95%, confirming the spatial symmetry of the simulated flow field.

3.6 Experimental–Numerical Comparison Procedure

Velocity data are reported in the in-plane coordinate system of velten2026: the horizontal in-plane axis ξ and the vertical streamwise axis z . Simulation planes are extracted at the physical positions corresponding to Pos 1 and Pos 3 ($a_{\text{Pos}} = -22.5$ and -7.5 mm from the reactor centre), and at the streamwise heights of layers L13–L17, matching the PIV measurement layers of velten2026. Each extracted plane is masked to the fluid region, interpolated onto the experimental PIV grid (vector resolution 3.40–3.44 vec/mm, Table 2 of velten2026), and normalised by the intrinsic mean velocity $v_{\text{int}} = v_s/\varphi$.

Write §3.6 – add error metrics definitions (RMSE, Pearson, mean relative difference)

3.7 Validation Results

Write §3.7 – validation results; insert Fig. 4 and Fig. 5

4 Flow Locking Framework

The central, novel contribution of this paper. Prior work velten2026 noted qualitatively that the flow “repeats across layers and is antisymmetric about the central axis.” Here that observation is turned into a measured quantity:

the mean flow repeats with the same twist as the geometry, and we score how strongly with a single order parameter that approaches unity for the periodic packings, collapses for the irregular packing, and weakens as the Reynolds number rises.

4.1 Physical Interpretation: Twist-and-Repeat Flow Locking

Write §4.1 – physical concept in plain language

4.2 Locking Order Parameter $L(n)$

The locking order parameter $L(n)$ is the cosine similarity between the de-rotated velocity fields of layer n and layer $n + N$:

$$L(n) = \frac{\langle \tilde{\mathbf{u}}_n \cdot \tilde{\mathbf{u}}_{n+N} \rangle}{\|\tilde{\mathbf{u}}_n\| \|\tilde{\mathbf{u}}_{n+N}\|}, \quad (18)$$

where $\tilde{\mathbf{u}}_n$ is the time-mean in-plane velocity field of layer n , de-rotated by $-n\theta$ to a common reference frame, and $N = 180^\circ/\theta$ is the layer period ($N = 6$ for $\theta = 30^\circ$; $N = 9$ for $\theta = 40^\circ$). $L = 1$ means the de-rotated layers are identical (perfectly locked); $L \rightarrow 0$ means no repeat. For the irregular packing (no period N) adjacent layers ($n \rightarrow n + 1$) are compared instead.

Write §4.2 – construction, pair-mask note, development length

4.3 Antisymmetry Index $A(n)$

Write §4.3 – antisymmetry index; revisit A definition if needed

4.4 Locking Across Topologies

Figure ?? shows the layer-to-layer locking matrix and $L(n)$ versus layer for the three topologies. For the 30° packing all off-diagonal entries are $L \approx 1.000$, confirming near-perfect locking across the developed window (L13–L17). The 40° packing locks at $L \approx 0.99$, still strongly locked but slightly reduced owing to the larger per-layer twist angle. The irregular packing yields $L = 0.10$ – 0.68 with no systematic pattern, confirming that the geometric twist-and-repeat rule is a necessary condition for flow locking.

Write §4.4 – quantify L for each topology; add development curve

4.5 Locking Versus Reynolds Number

Write §4.5 – Re effect on L; insert Fig. 7d (Re trend panel)

4.6 Physical Interpretation and Hand-off to Paper 2

Write §4.6 – physical interpretation; forward pointer to Paper 2

5 Flow Organisation

5.1 Mean Flow Picture

Write §5.1 – mean flow picture

5.2 Channelling and Transverse Spreading

Write §5.2 – channelling index and transverse KE fraction

5.3 Recirculation and Secondary Motion

Write §5.3 – recirculation; decide on optional vortex diagnostics

5.4 Topology Contrast: 30° vs 40° vs Irregular

Write §5.4 – topology contrast; insert Figs. 9 and 10

6 Conclusions

Write Conclusions

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